NASA SPACE APPS 2026

SoilSight

Earth data. Farm wisdom.

An explainable crop-rotation and climate-adaptation planner that turns NASA Earth data into field-level planning conversations.

FIELD SHIFT **Adapting Farms with NASA Data**

Independent challenge prototype · November 2026

Farm decisions are local. Climate data is not.

Farmers must act before uncertainty becomes visible in the field.

1 CLIMATE VOLATILITY

Changing heat and rainfall shift sowing windows, water demand and seasonal risk.

2 FRAGMENTED EVIDENCE

NASA observations, soil estimates and crop requirements live in separate technical systems.

3 COMPETING PRIORITIES

Water, soil health, market value and resilience rarely point to the same rotation.

4 TRUST GAP

A score without sources, assumptions or explanations is difficult to act on.

THE DESIGN QUESTION **How might we turn regional Earth data into an understandable, farmer-controlled crop-rotation decision?**

What the Field Shift challenge asks us to connect

The solution must be a decision-support tool—not simply another agricultural dashboard.

1. **NASA Earth observations** — Regional environmental context
2. **Local soil information** — Field conditions and limitations
3. **Crop characteristics** — Growing windows and agronomic traits
4. **Farmer priorities** — Water, soil, market and resilience

THE REQUIRED COMPARISON

At least three rotation strategies—including the farmer's existing practice.

SoilSight now evaluates editable current practice beside Soil-first and Water-saver alternatives.

CORE NOW **NASA POWER · MODIS via GIBS**

ROADMAP SMAP · GPM · HLS · ECOSTRESS · GRACE · SRTM · GEDI

From a drawn boundary to an explainable plan

One continuous workflow keeps data, decisions and uncertainty connected.

1 **DRAW** — **Field boundary**

2 **LOAD** — **NASA climate**

3 **SCREEN** — **12-month risk**

4 **COMPARE** — **Three rotations**

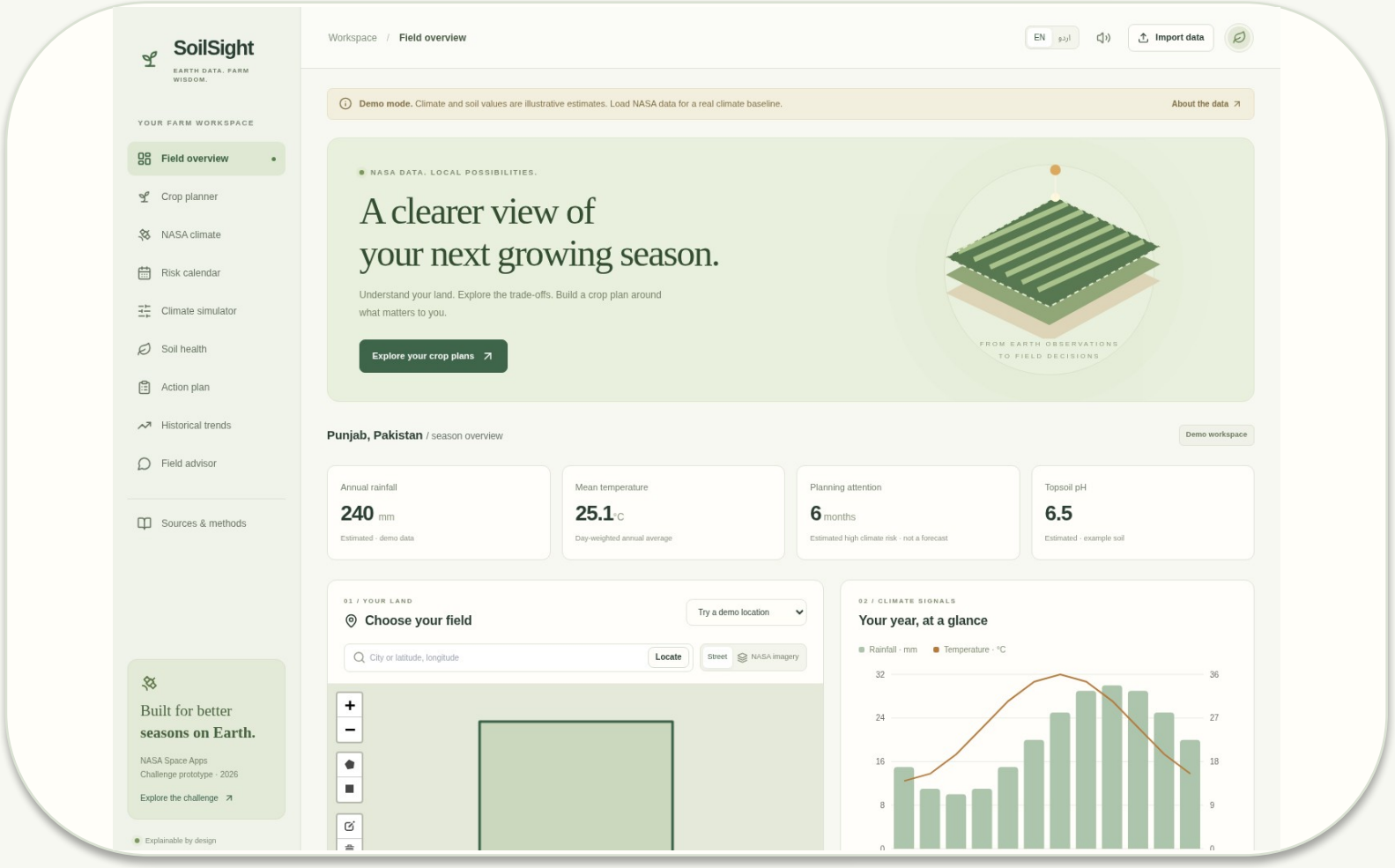
5 **WEIGH** — **Farmer priorities**

6 **STRESS** — **Hotter / drier**

7 **ACT** — **Season plan**

OUTPUT A ranked rotation, monthly water gaps, plain-language “why”, assumptions, and an exportable action plan.

A farm workspace built around one selected field



FIELD CONTEXT

The farmer draws or edits a polygon. Its area is calculated, and its centre selects the regional NASA grid cell.

VISIBLE AT A GLANCE

- Annual rainfall
- Mean temperature
- Months needing attention
- Estimated topsoil pH

Demo values remain visibly labelled until live NASA data is loaded.

NASA resources: what drives the score—and what does not

A precise boundary between implemented evidence and future layers protects scientific credibility.

CORE QUANTITATIVE SOURCE

NASA POWER

- Temperature: T2M_MIN / MAX / MEAN
- Precipitation: PRECTOTCORR
- Wetness indices: GWETROOT / GWETPROF
- 15 complete years for descriptive trends

Regional model grid ≈ 0.5° × 0.625°

CONTEXTUAL MAP

MODIS Terra imagery via NASA GIBS. Visible behind the boundary; never used to calculate a recommendation.

COMPLEMENTARY INPUT

ISRIC SoilGrids estimates for 0–30 cm plus an illustrative crop library. Both labelled non-NASA and estimated.

CHALLENGE RESOURCE ROADMAP

SMAP / ECOSTRESS	Water stress
GPM IMERG	Rainfall detail
Landsat / HLS	Cover history
GRACE-FO	Groundwater trend
SRTM / GEDI	Terrain / structure

Future datasets are disclosed as roadmap items—not silently implied in today's score.

A 12-month risk calendar, not a weather forecast

Monthly climate patterns become planning attention signals.

J F M A M J J A S O N D

COLD STRESS Monthly minimum temperatures

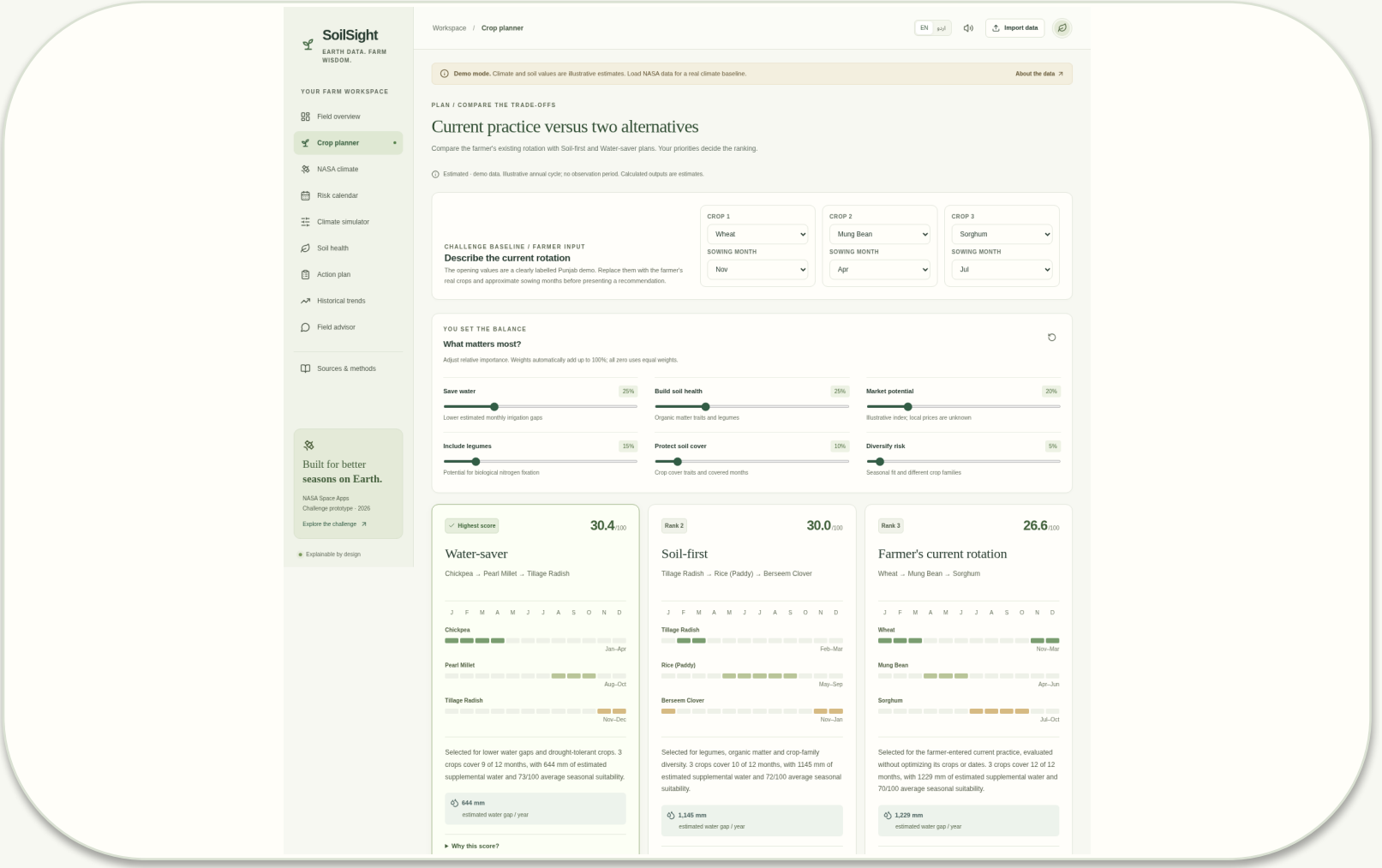
HEAT STRESS Monthly maximum temperatures

RAINFALL DEFICIT Low monthly precipitation

EXCESS RAIN High monthly precipitation

IMPORTANT **Monthly averages cannot predict flooding, frost events or next week's weather.**

Current practice versus two generated alternatives



1 CURRENT PRACTICE

Editable crops + sowing months. Evaluated as entered.

2 SOIL-FIRST

Rewards legumes, organic-matter traits and family diversity.

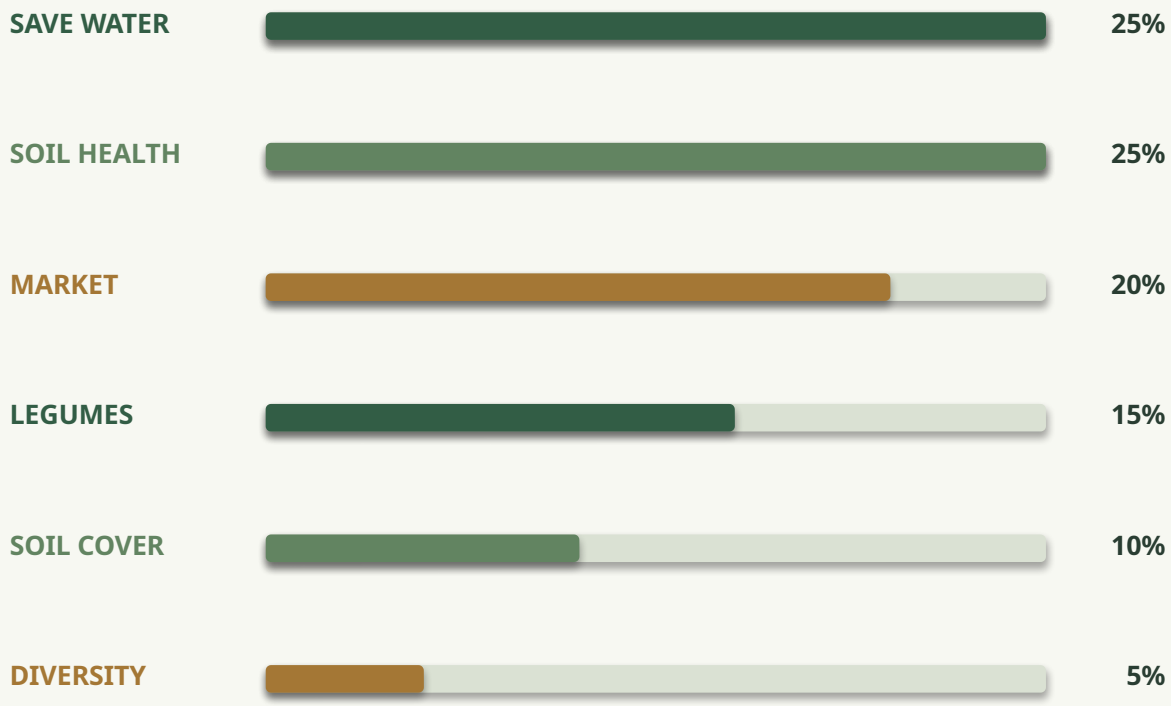
3 WATER-SAVER

Favors drought tolerance and lower monthly water gaps.

The farmer’s priorities rank all three using the same transparent score.

Every ranking can be taken apart

Six user-controlled priorities become normalized weights and visible score contributions.



EXAMPLE SCORE TRACE

Water conservation **82.0**

× 25% normalized weight

20.5 points

The displayed overall score is the actual weighted sum used to rank plans—not a success probability.

All-zero priorities → equal weights. Every criterion, weight and contribution remains visible.

What if the same plan faces a hotter, drier season?

The simulator holds crops and dates fixed, then recalculates suitability, ET_0, water gaps and rank.

MONTHLY WATER BALANCE

Crop need = $ET_0 \times Kc \times$ stage factor

Water gap = max(0, need − 80% rain)

Field volume = water gap × area

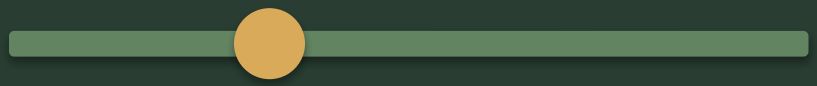
Planning estimate—not an irrigation schedule.

SENSITIVITY CONTROLS

Temperature **+2.0°C**



Rainfall **−20%**



Not an IPCC scenario or forecast. It answers: “Does this exact plan remain competitive under changed assumptions?”

Every figure says what kind of evidence it is

Confidence labels travel with the data, calculations and exported plan.

OBSERVED

Direct measurements with documented provenance.

Currently reserved—no field-observation claim.

MODELLED

Live NASA POWER regional reanalysis.

Not a field sensor or weather forecast.

ESTIMATED

SoilGrids, crop traits, scores and scenarios.

Requires local validation before action.

EXPERT REVIEW BANNER

Validate cultivars, planting windows, water access, prices and a current soil test with a local agronomist.

A deployable browser application with no API secrets

Static architecture reduces cost, deployment risk and demo-day failure points.

INTERFACE	**React · TypeScript · responsive · Urdu / RTL**
GEOSPATIAL	**Leaflet · drawing · spherical area · centroid**
DECISION ENGINE	**Seasonal screening · scoring · irrigation · stress**
PUBLIC DATA	**NASA POWER · GIBS · SoilGrids · OSM / Nominatim**

VERIFIED BUILD

44 calculation & data tests

8 browser journeys

0 known npm vulnerabilities

Static dist/ deploy · no backend required

SoilSight continues where a dashboard usually stops

The innovation is the connection between evidence, trade-offs and farmer agency.

1 **CONTEXT**

A real field boundary anchors every query.

2 **EVIDENCE**

NASA climate becomes seasonal signals.

3 **CHOICE**

Priorities change the result transparently.

4 **EXPLANATION**

Every rank exposes its contributions.

5 **ADAPTATION**

The same plan can be stress-tested.

6 **ACTION**

The winner becomes a portable season plan.

NASA sees the environment. Farmers define what matters. SoilSight makes the trade-off visible.

From challenge prototype to locally validated tool

A focused roadmap deepens evidence without hiding uncertainty.

1 **NOW** — POWER climate · current rotation · 3-way comparison · water gaps · stress test

2 **VALIDATE** — Local agronomist review · cultivar windows · soil tests · market assumptions

3 **ENRICH** — SMAP / GPM / ECOSTRESS validation · SRTM erosion · HLS field-cover history

4 **REACH** — Complete Urdu review · voice testing · low-bandwidth deployment · farmer trials

SUCCESS MEASURE · A farmer can explain why one rotation fits their priorities better than another.

SOILSIGHT

A planning conversation, before a planting decision.

NASA Earth data + local knowledge + farmer priorities

THE PROMISE

Explainable by design. Honest about uncertainty. Built for better seasons on Earth.

Thank you

Primary sources and methods

URLs remain editable and clickable in the presentation.

Official Field Shift resources	spaceappschallenge.org/2026/challenges/adapting-farms-with-nasa-data/?tab=resources
NASA POWER API	power.larc.nasa.gov/docs/services/api/
NASA GIBS	nasa-gibs.github.io/gibs-api-docs/
NASA SMAP / GPM	smap.jpl.nasa.gov/data/ · gpm.nasa.gov/data
NASA HLS / ECOSTRESS	earthdata.nasa.gov/data/projects/hls · ecostress.jpl.nasa.gov
NASA GRACE / SRTM / GEDI	grace.jpl.nasa.gov/data · earthdata.nasa.gov/data/instruments/srtm · gedi.umd.edu
ISRIC SoilGrids	docs.isric.org/globaldata/soilgrids/
FAO-56 / EcoCrop context	fao.org/4/x0490e/x0490e00.htm · gaez.fao.org/pages/ecocrop

NASA supplies data; it has not validated or endorsed SoilSight's crop recommendations.